Lucas Toñanez

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EDUCATION

University of Central Florida **Orlando, FL**

Bachelor of Science in Computer Science *Expected May 2027*

- GPA: 3.4/4.0 | Bright Futures Scholarship | Millican Gold Merit Award
- Relevant Coursework: Artificial Intelligence, Algorithms for Machine Learning, Data Structures & Algorithms, Software Development

PUBLICATIONS

Conditional Multi-Event Temporal Grounding in Long-Form Video — Yuanhao Zou, Arthad Kulkarni, **Lucas Toñanez**, et al. arXiv:2606.15320, 2026 (Computer Vision).

RESEARCH EXPERIENCE

UCF Institute for Artificial Intelligence (IAI) **Orlando, FL**

Undergraduate Research Assistant — Video Understanding / Computer Vision *Dec 2025–Present*

- Contributed to and integrated components of an **adaptive query planner and query-aware routing pipeline** for CoMET-style long-video temporal grounding across candidate retrieval, localization, and refinement stages.
- Integrated the **end-to-end multimodal temporal-grounding pipeline**; in one profiled run, query-window clipping/caching reduced scene-detection workload from **307 to 91 candidate segments (~70%)**, with batched inference and CUDA resource management improving repeatability and throughput.
- Built **evaluation and benchmark infrastructure** for JSONL prediction extraction, runtime/error tracking, checkpoint/resume, and consistency validation across large experiment runs.
- Reproduced, integrated, and debugged **Qwen, VideoMind, and DoraemonGPT baselines**; compared temporal predictions and documented model behavior and failure modes.
- Developed a **600+ video query-generation and data workflow** using Qwen, Gemini, Python, FFmpeg, OpenCV, and structured JSON; also contributed confidence-aware temporal refinement for revisiting candidate windows.

Undergraduate Researcher — Self-Evolving Video Annotation *Sep 2026–Present*

- Built a pinned Qwen3-VL-8B Annotator/Corrector video-annotation pipeline with a Qwen2.5-VL baseline, frozen-draft evaluation, hash-verified inputs, and 100+ CPU contract tests.
- Audited the temporal-grounding Corrector and designed a frame-cited boundary-evidence protocol; in a bounded four-call pilot on one reviewed clip, it avoided the baseline's harmful edit but made no corrective edits.

Undergraduate Researcher — Vision-Language-Action / Robotics *Jun 2026–Present*

- Worked with **OpenPI pi0.5 and LeRobot** to record policy rollouts, support real-robot deployment, and collect positive and negative task feedback.
- Mapped the proposed reuse/replan → training/update → redeployment pipeline and evaluated feedback-labeling requirements, identifying annotation scalability as a constraint before full-loop implementation.

Undergraduate Researcher — Few-Shot Esports Event Spotting (Dota 2) *Oct 2026–Present*

- Built a few-shot event-spotting benchmark on Dota 2 gameplay video with frozen CLIP and R3D-18 features, replay-log-aligned labels, temporal context features, background rejection, and a tested ±0.5/±1 s event-mAP evaluator on sealed, match-disjoint holdouts.
- On four unused matches, a validation-tuned background penalty raised Mars event mAP (±1 s) from 0.247 to 0.293 with gains on both matches; documented the remaining false alarms (~7/min) and weak transfer to other ability classes.

SELECTED PROJECTS

Gasoline Price Movement Classification **PyTorch, NumPy**

- Built a leakage-safe pipeline over 142,818 French station–fuel sequences and compared MLP, 1D CNN, LSTM, and Transformer models via an 80-trial search with 3-seed reruns; Transformer reached 0.666 validation macro-F1 vs. 0.635 for logistic regression.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS | AI/ML: PyTorch, Hugging Face, Qwen, DINOv2, RAFT, OpenPI pi0.5, LeRobot

Systems & Data: ROS2, CUDA, Linux, FFmpeg, OpenCV, JSON, Git | **Research:** multimodal AI, temporal grounding, long-video understanding, VLA/robotics, evaluation pipelines

ADDITIONAL PROFESSIONAL EXPERIENCE

Tecalemit Inc. — Software Engineering Intern, AI & Automation **Sep 2024–2025**

- Built internal software tools and REST API integrations; prototyped AI-assisted customer-service and sales workflows using OpenAI APIs, structured product context, and email handoffs.
- Tested, debugged, and documented behavior across local and deployed environments using Git, Postman, and lightweight server tooling.

UCF Institute for Artificial Intelligence — Student Assistant **2025–Present**

- Support research and education initiatives through concise project coordination, meeting preparation, and documentation.